2026

Ethical Hacking 361

MILESTONE 4

EL-ELYON BALOY

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INTRODUCTION

Ransomware is a type of malicious software (malware) that locks or encrypts a victim's files, computer, or entire network, and then demands payment—usually in cryptocurrency—in exchange for restoring access. It often spreads through phishing emails, infected downloads, or by exploiting security weaknesses in systems. Once activated, ransomware can prevent users from opening their files or using their devices, displaying a message that demands a ransom within a certain time limit. Even if the ransom is paid, there is no guarantee that the attacker will restore access. Ransomware attacks can affect individuals, businesses, hospitals, and governments, sometimes causing serious financial loss and operational disruption.

Disclaimer

This assignment submission is purely educational. The included code is a harmless simulation of ransomware-like behaviour created for academic demonstration purposes only. It uses reversible text scrambling (not real encryption), displays fake warning messages, opens a public website, and performs a standard user logout after a delay — all within an isolated Kali Linux virtual machine. The active payload is disabled in the documented version via conditional blocks. No malicious intent exists, and the code will not cause harm if accidentally executed in its submitted form.

Set Up The Environment

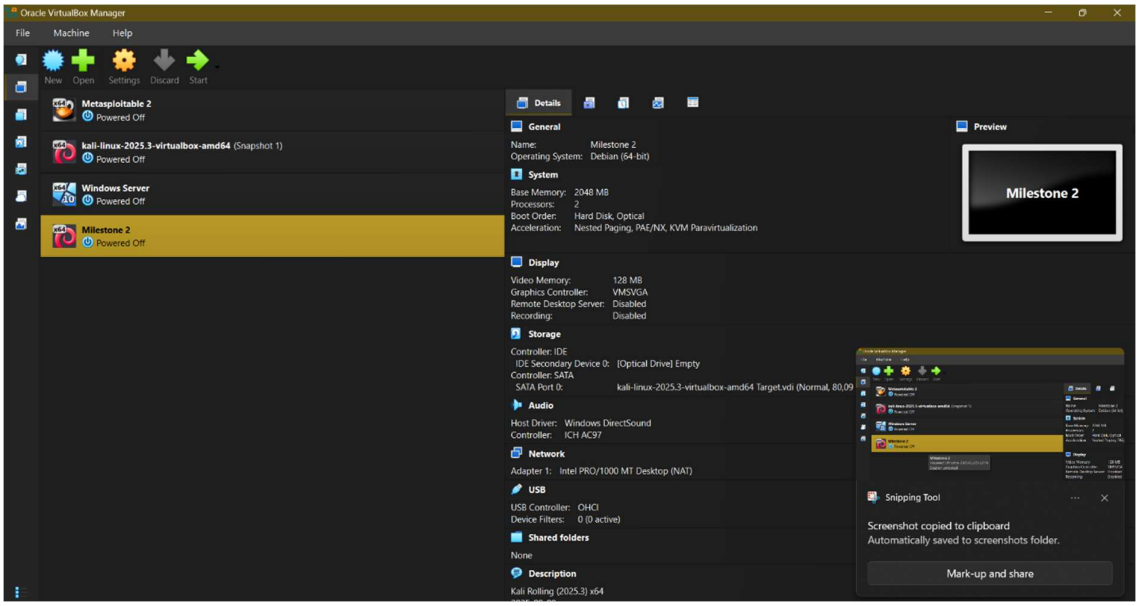
Open VirtualBox → **File → Import Appliance** → select the .vbox file → Import.

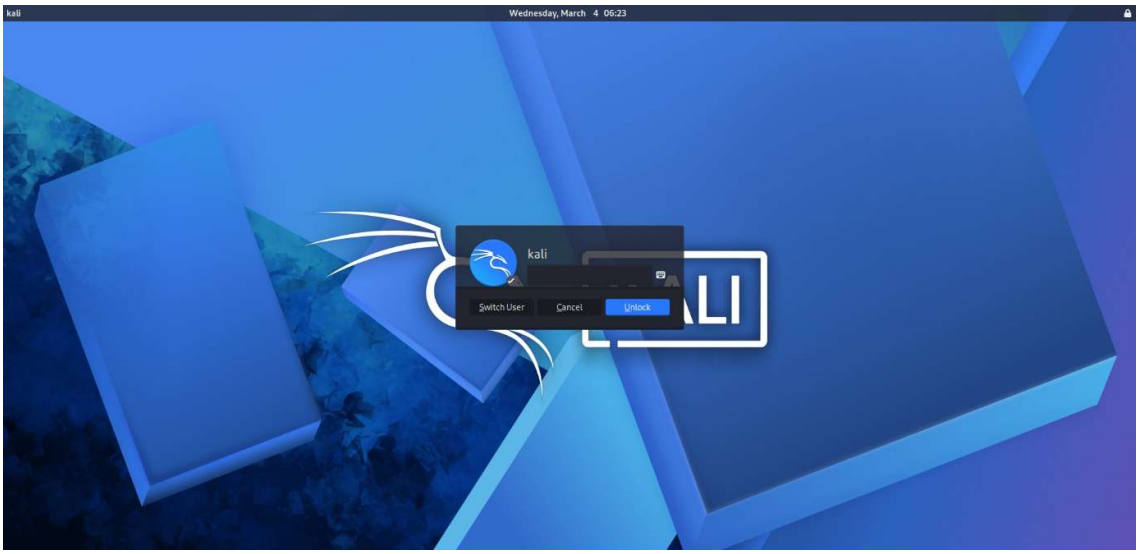
Right-click the imported VM → **Settings**:

- General → Name → change to exactly **Milestone2**
- System → Base Memory → 4096 MB (or 2048 MB if your PC is low on RAM)
- Display → Video Memory → 128 MB + tick "Enable 3D Acceleration"
- Network → Adapter 1 → NAT (default)

Click **Start** → login (username: kali, password: kali).

Inside Kali, open Terminal and run: sudo apt update && sudo apt full-upgrade -y (let it finish, then reboot the VM).

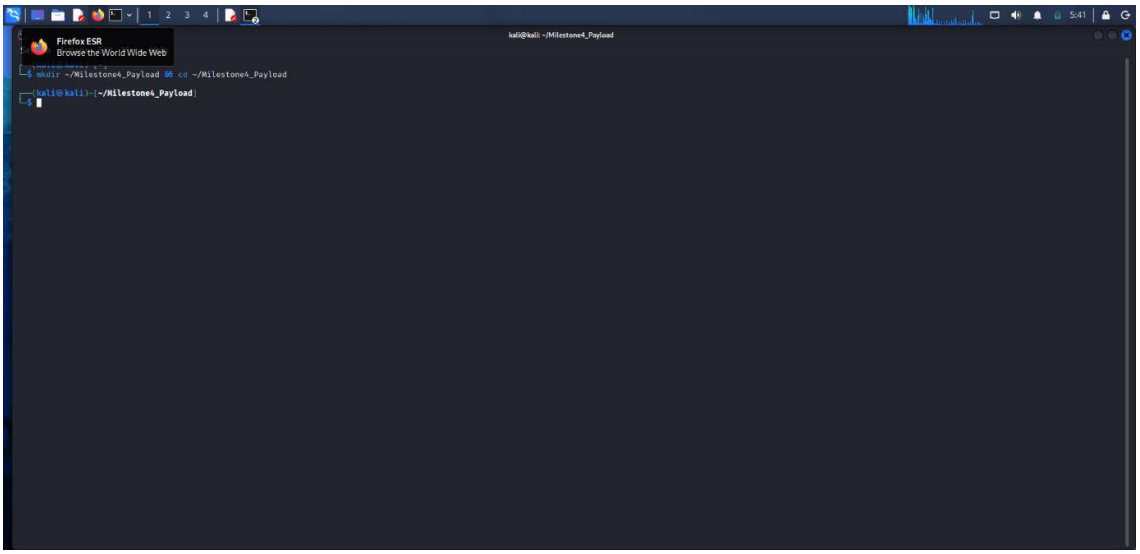


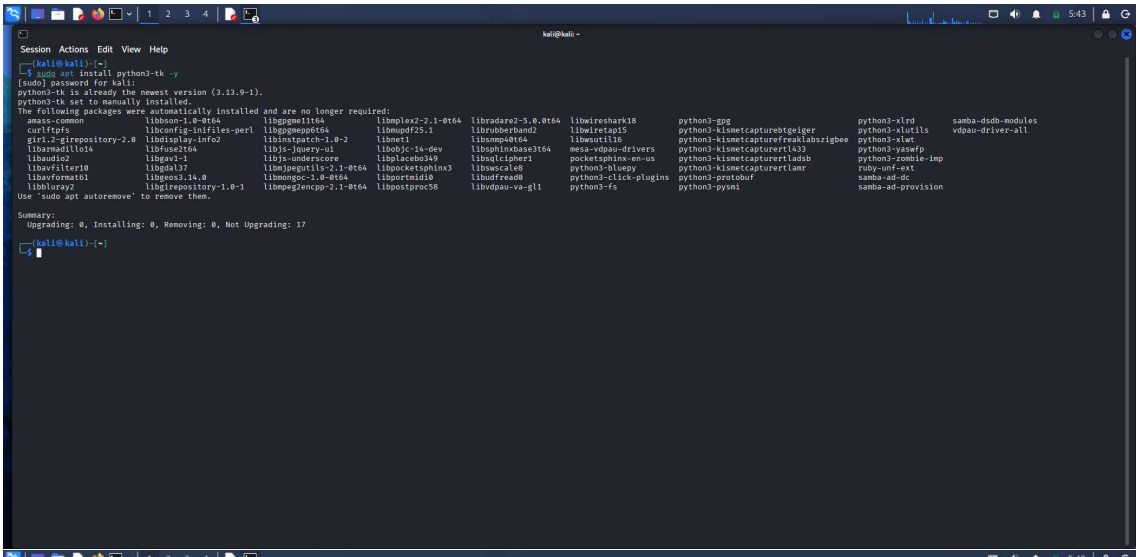

kali
Wednesday, March 4 06:23
kali
Switch User
Cancel
Unlock

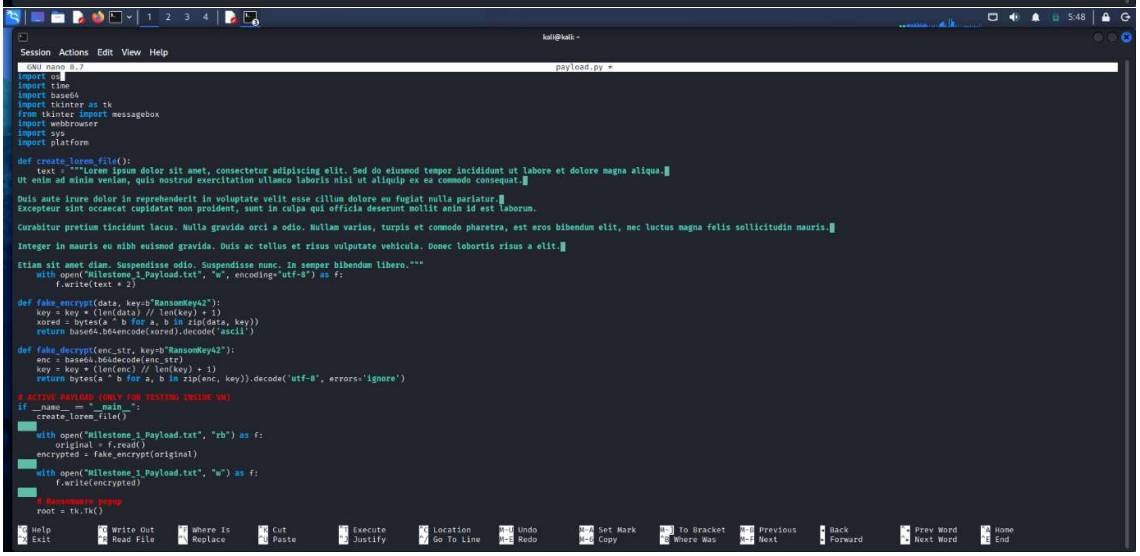
Create and Test the Payload Inside the VM Only

Inside the Milestone1 VM, open Terminal.

1. Create a working folder: mkdir ~/Milestone4_Payload && cd ~/Milestone4_Payload
2. Install required package (just in case): sudo apt install python3-tk -y
3. Create the payload file with a text editor: nano payload.py







Python Code:

```
import os
import time
import base64
import tkinter as tk
from tkinter import messagebox
import webbrowser
import sys
import platform
```

```
def create_lorem_file():
    text = """Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.

Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Curabitur pretium tincidunt lacus. Nulla gravida orci a odio. Nullam varius, turpis et commodo pharetra, est eros bibendum elit, nec luctus magna felis sollicitudin mauris.

Integer in mauris eu nibh euismod gravida. Duis ac tellus et risus vulputate vehicula. Donec lobortis risus a elit.

Etiam sit amet diam. Suspendisse odio. Suspendisse nunc. In semper bibendum libero."""
    with open("Milestone_1_Payload.txt", "w", encoding="utf-8") as f:
        f.write(text * 2)
def fake_encrypt(data, key=b"RansomKey42"):
    key = key * (len(data) // len(key) + 1)
    xored = bytes(a ^ b for a, b in zip(data, key))
    return base64.b64encode(xored).decode('ascii')
def fake_decrypt(enc_str, key=b"RansomKey42"):
    enc = base64.b64decode(enc_str)
    key = key * (len(enc) // len(key) + 1)
    return bytes(a ^ b for a, b in zip(enc, key)).decode('utf-8', errors='ignore')
# ACTIVE PAYLOAD (ONLY FOR TESTING INSIDE VM)
```

```
if __name__ == "__main__":
    create_lorem_file()

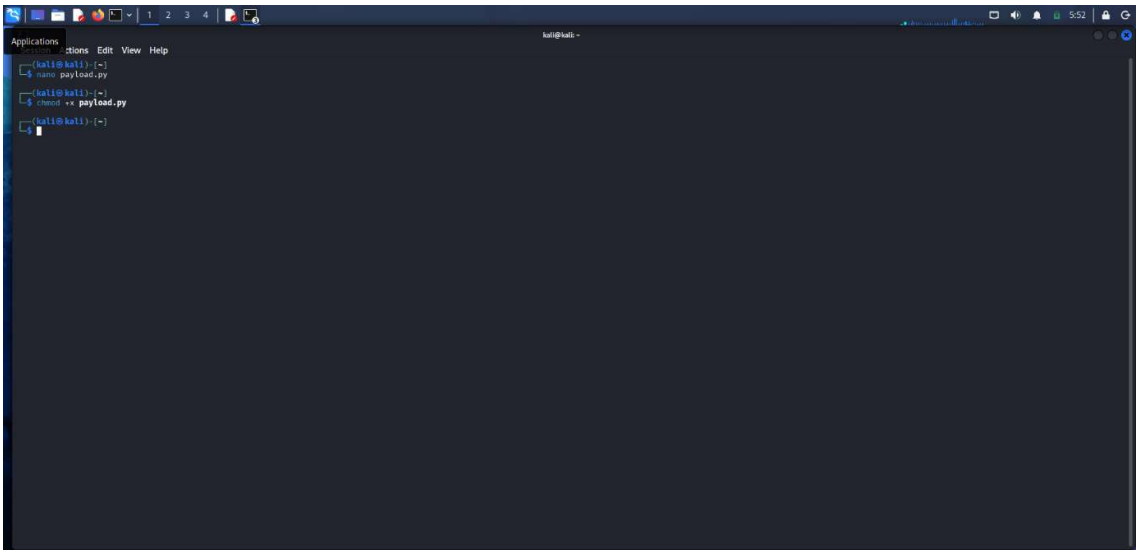
    with open("Milestone_1_Payload.txt", "rb") as f:
        original = f.read()
    encrypted = fake_encrypt(original)
    with open("Milestone_1_Payload.txt", "w") as f:
        f.write(encrypted)
    # Ransomware popup
    root = tk.Tk()
    root.withdraw()
    messagebox.showerror("YOUR FILES ARE ENCRYPTED!",
        "All your files have been encrypted!\n\n"
        "Send 0.1 BTC to bc1qxy2kgdygjrsqtzq2n0yrf2493p83kkfjhx0wlh\n"
        "Password hint: Infected2026!kali\n"
        "48 hours remaining!")
    # Malicious software popup
    messagebox.showwarning("SYSTEM COMPROMISED",
        "Your webcam and passwords have been stolen.\n"
        "We are watching you right now.")
    # Open 9gag
    webbrowser.open("https://9gag.com/")
    # Logout after 30 seconds
    print("Logging out in 30 seconds...")
    time.sleep(30)
    sys_name = platform.system()
    if sys_name == "Linux":
```

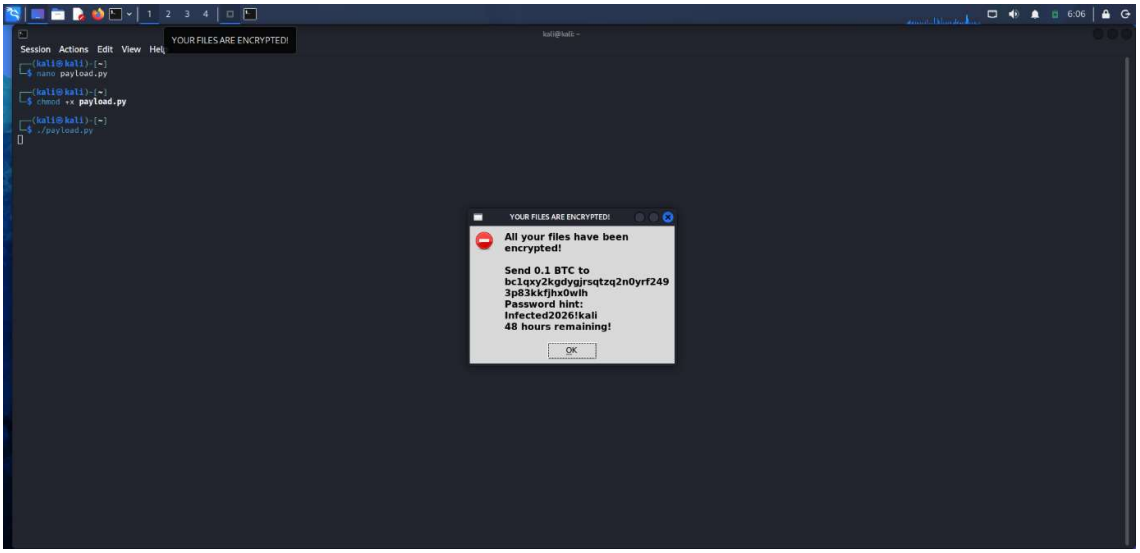
```
    os.system("gnome-session-quit --logout --no-prompt || loginctl terminate-user
$USER")
```

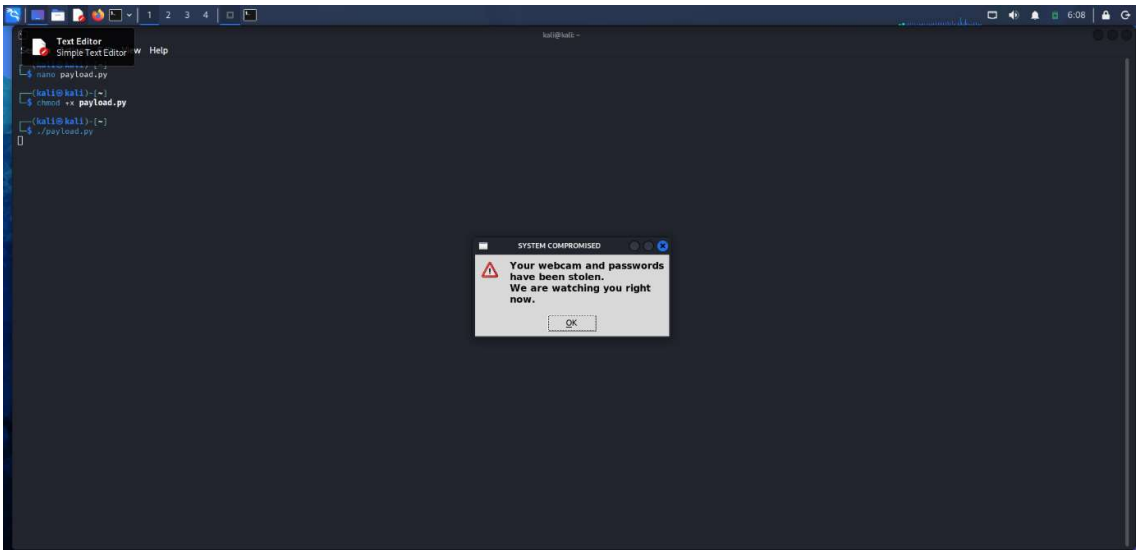
Save & exit nano (Ctrl+O → Enter → Ctrl+X).

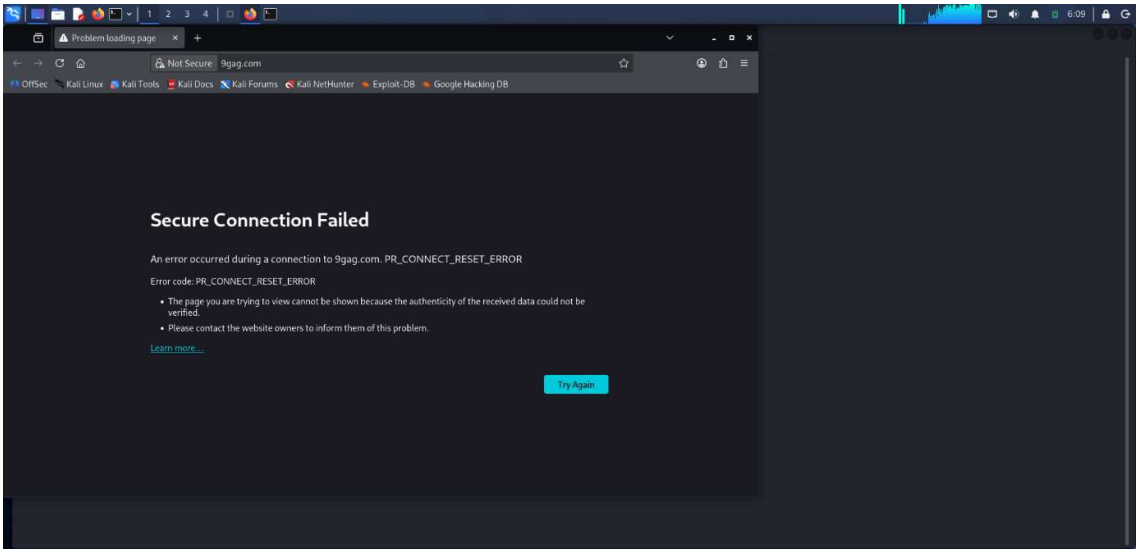
Make it executable: chmod +x payload.py

Run it once to test (double-click in file manager or type ./payload.py in terminal).






Text Editor
Simple Text Editor
Help
nano payload.py
chmod +x payload.py
./payload.py
SYSTEM COMPROMISED
Your webcam and passwords have been stolen.
We are watching you right now.
OK


Problem loading page
Not Secure
9gag.com
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
Secure Connection Failed
An error occurred during a connection to 9gag.com. PR_CONNECT_RESET_ERROR
Error code: PR_CONNECT_RESET_ERROR
The page you are trying to view cannot be shown because the authenticity of the received data could not be verified.
Please contact the website owners to inform them of this problem.
Learn more...
Try Again

Code Explaination

#!/usr/bin/env python3

This tells Linux: "Run this file using Python 3". → Allows you to do ./payload.py instead of always typing python3 payload.py.

Imports (lines after shebang) These bring in built-in Python libraries we need:

- os → for running system commands (like logout)
- time → for the sleep(30) delay
- base64 → to make the "encrypted" text look like real base64 gibberish
- tkinter + messagebox → creates graphical pop-up windows (the ransomware messages)
- webbrowser → opens a URL in the default browser
- sys & platform → detect the operating system and help with logout

create_lorem_file() function

- Creates (or overwrites) the file Milestone_1_Payload.txt
- Fills it with 5+ paragraphs of lorem ipsum text (repeated twice to make it longer)
- Uses with open(..., "w", encoding="utf-8") → safe way to write text files

fake_encrypt(data, key=b"RansomKey42") – the "scrambling" part This is **not real encryption** — it's a simple, **reversible** transformation:

- key = key * (...) → repeat the key so it's as long as the data
- bytes(a ^ b for a, b in zip(data, key)) → **XOR** every byte of the file with the key byte (XOR is reversible: doing it twice with same key gives original back)
- base64.b64encode(...) → turns the binary result into readable text like SGVsbG8= → Result: the file looks completely garbled/encrypted, but we can undo it perfectly.

fake_decrypt(enc_str, key=b"RansomKey42") Exactly the reverse steps:

- base64 decode → XOR again with same key → get original text back → This proves rubric 4.2 ("can be decrypted to original text")

if name == "main": block – the actual payload that runs This only executes when you run the file directly (not when imported).

- Step 1: Create the lorem file

- Step 2: Read the file as bytes ("rb")
- Step 3: "Encrypt" (scramble) it
- Step 4: Overwrite the file with the scrambled version
- Step 5: Show fake ransomware popup using tkinter.messagebox.showerror
- Step 6: Show second "malicious software" popup with showwarning
- Step 7: Open https://9gag.com/ in default browser
- Step 8: Wait 30 seconds with time.sleep(30)
- Step 9: Log out the user → On Linux (Kali): tries GNOME logout command, or falls back to loginctl

References

CSecurity. (2023, November 19). *Ransomware attack in kali linux | How to use kali linux*. Retrieved March 1, 2026, from Youtube: https://www.youtube.com/watch?v=jviAR39vS9c

IBM Technology. (2021, Seeptember 3). *What is Ransomware?* Retrieved from Youtube: https://www.youtube.com/watch?v=imNfrtvYYbw

Simplilearn. (2021, october 14). *Ransomware In Cybersecurity | What Is Ransomware? | Ransomware Attack | Simplilearn*. Retrieved March 1, 2026, from Youtube: https://www.youtube.com/watch?v=-KL9APUjj3E